

Triangulation Computations

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Field Exercise

Aim:

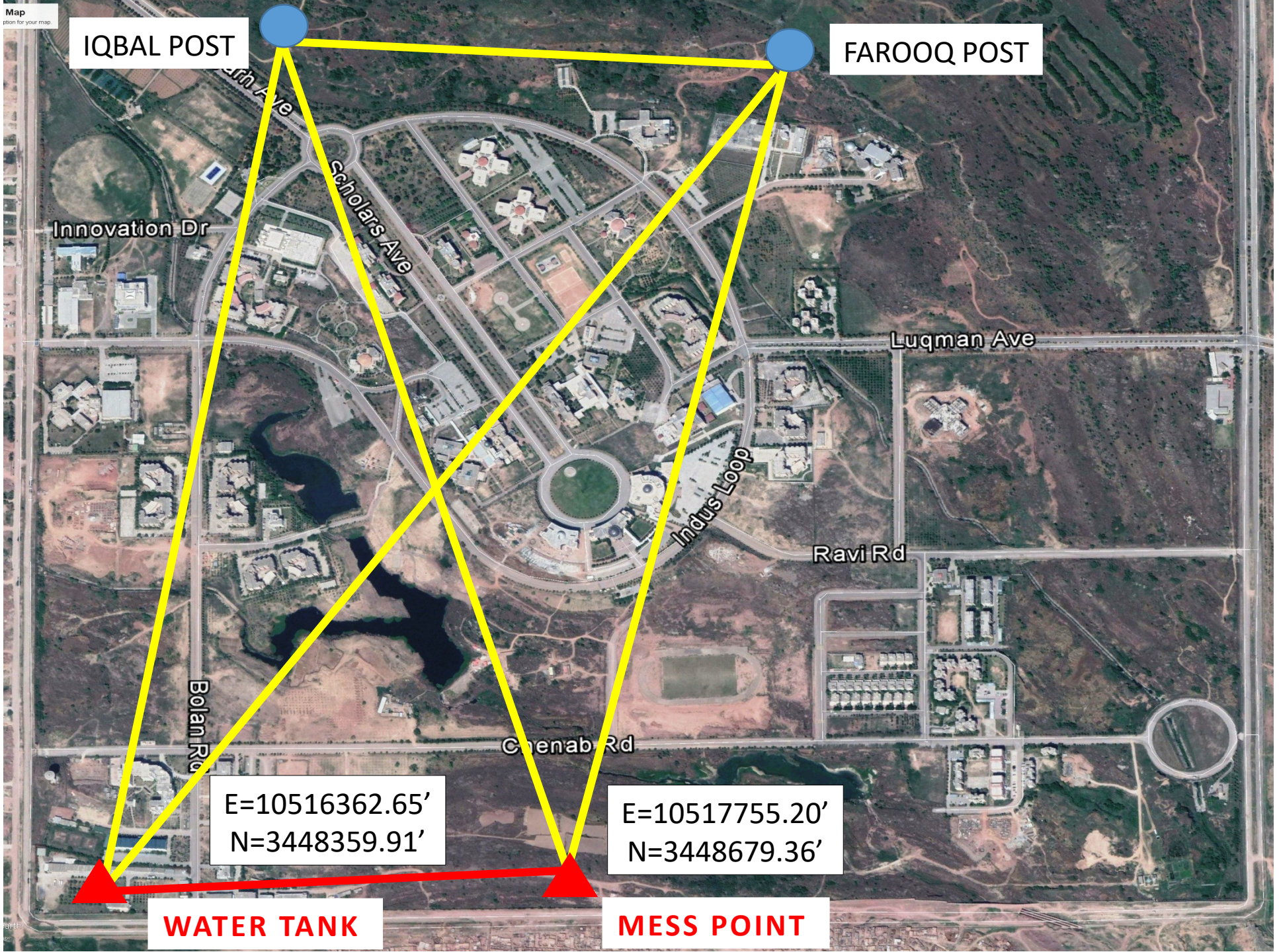
Determine the internal horizontal angles between Triangulation Points.

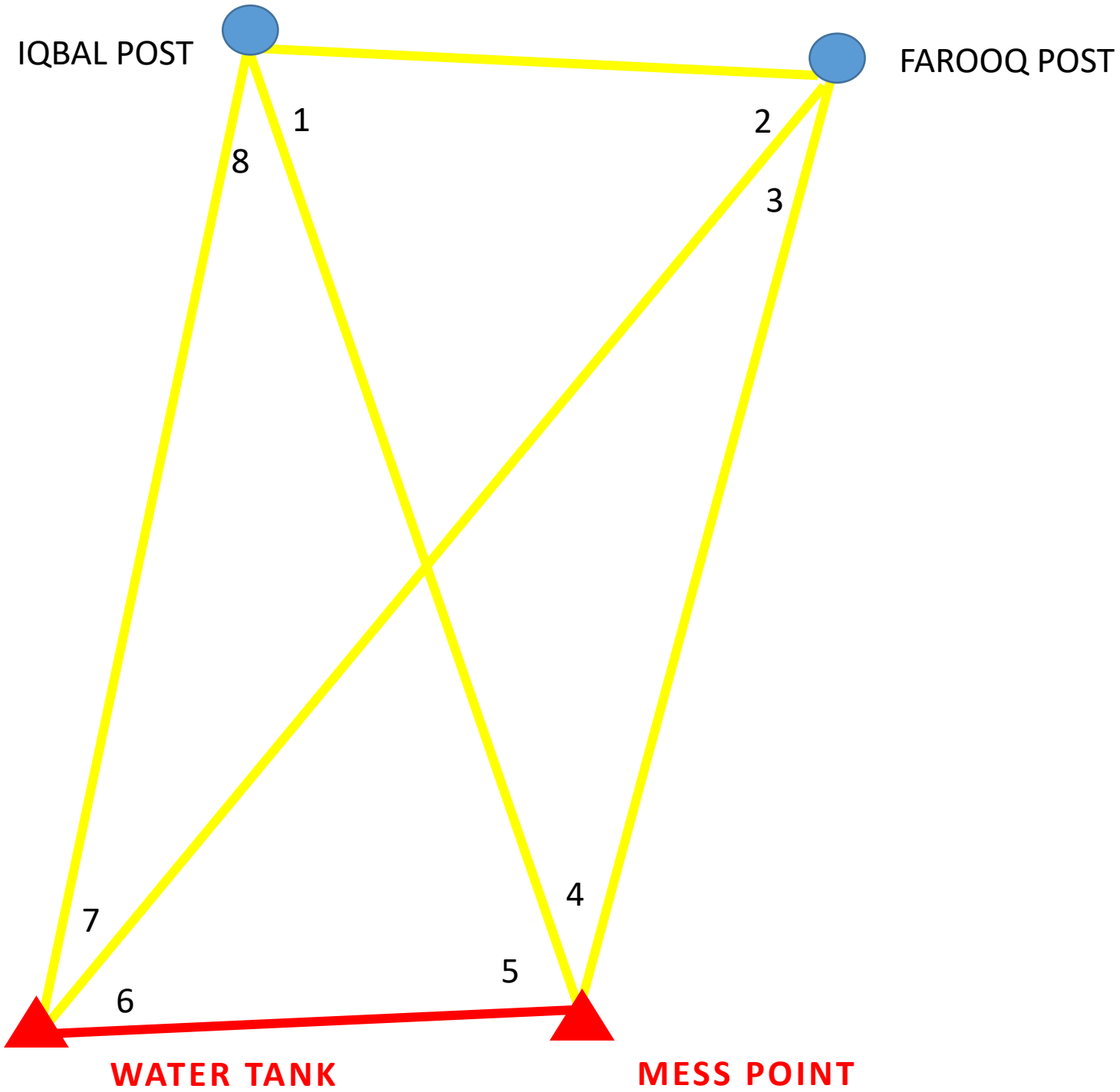
Equipment:

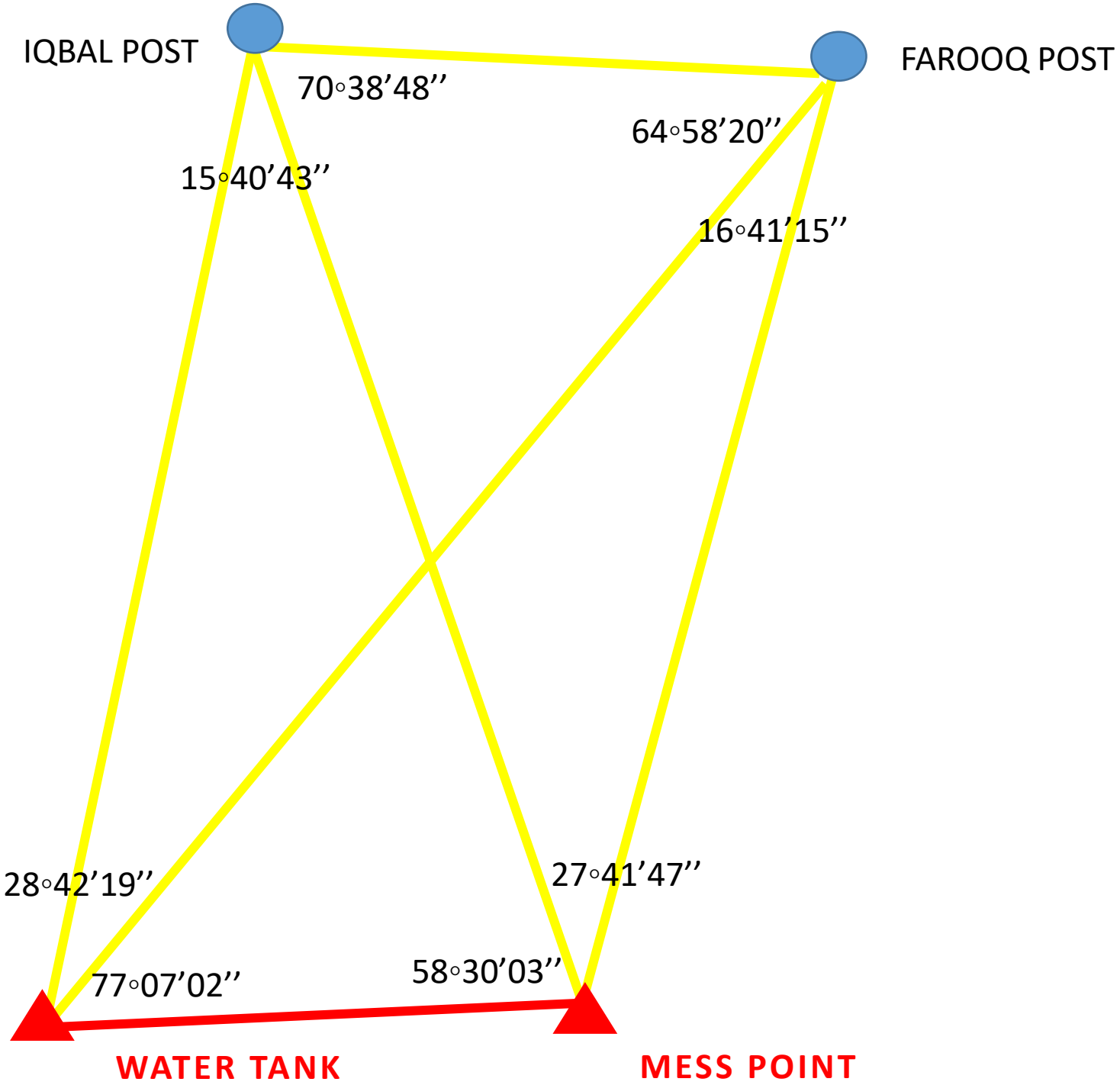
Total Station

Given Data:

Coordinates of Water Tank and Mess Point.


IQBAL POST
FAROOQ POST
Innovation Dr
Scholars Ave
Luqman Ave
Indus Loop
Ravi Rd
Bolan Rd
Chenab Rd
E=10516362.65’
N=3448359.91’
E=10517755.20’
N=3448679.36’
WATER TANK
MESS POINT


IQBAL POST
FAROOQ POST
1
2
3
4
5
6
7
8
WATER TANK
MESS POINT


IQBAL POST
FAROOQ POST
70◦38’48’’
64◦58’20’’
15◦40’43’’
16◦41’15’’
28◦42’19’’
27◦41’47’’
77◦07’02’’
58◦30’03’’
WATER TANK
MESS POINT

Computations

Step 1 : Angle Adjustment

Check 1

∑ Internal Angles = 360◦

Check 2

Angle 1 + Angle 2 = Angle 6 + Angle 5
Angle 7 + Angle 8 = Angle 3 + Angle 4

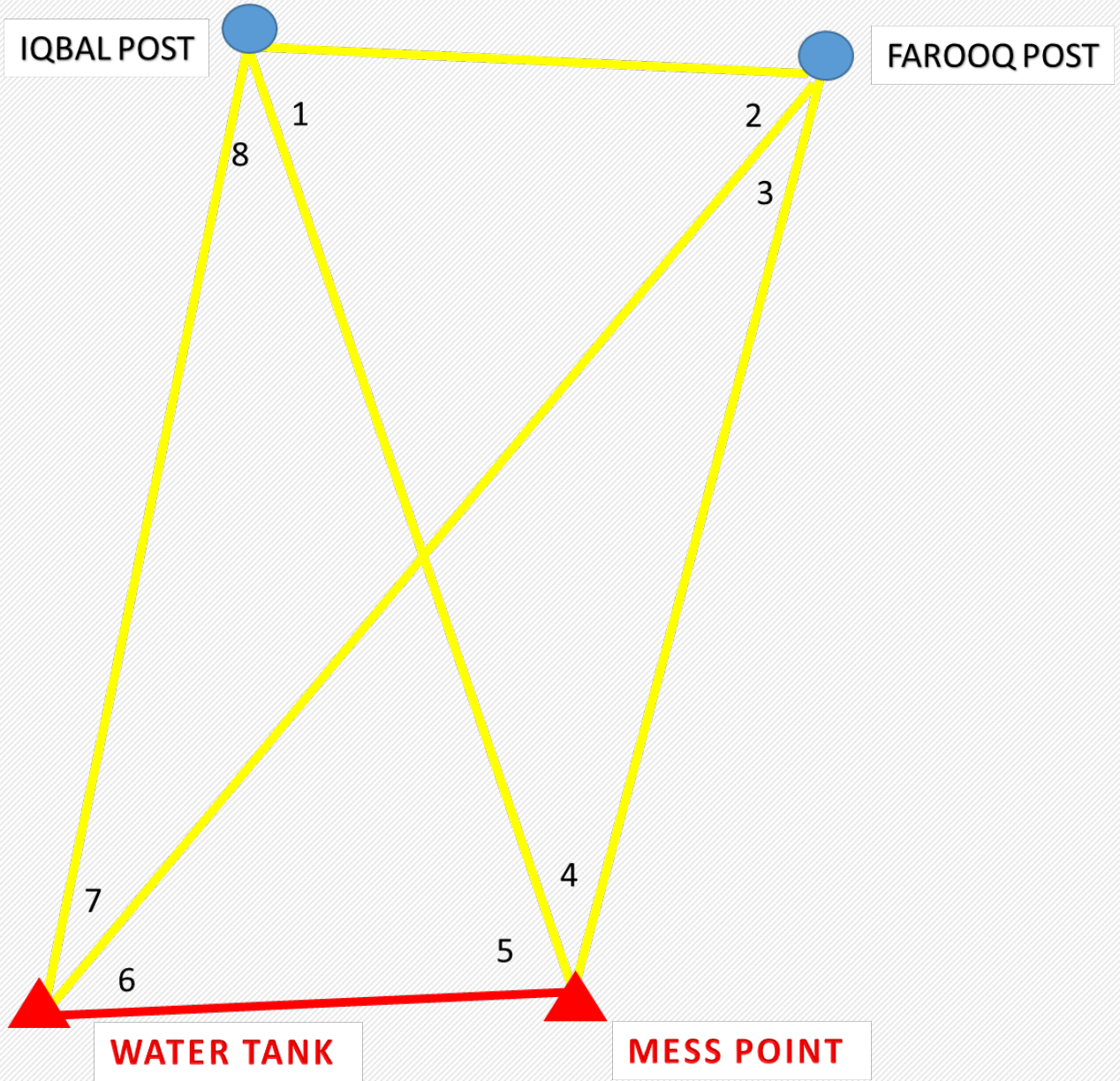
Check 3

∑ Internal Angles of Triangles = 180 ◦

Step 2 : Bearings

Step 3 : Distances

Step 4 : Co-ordinates



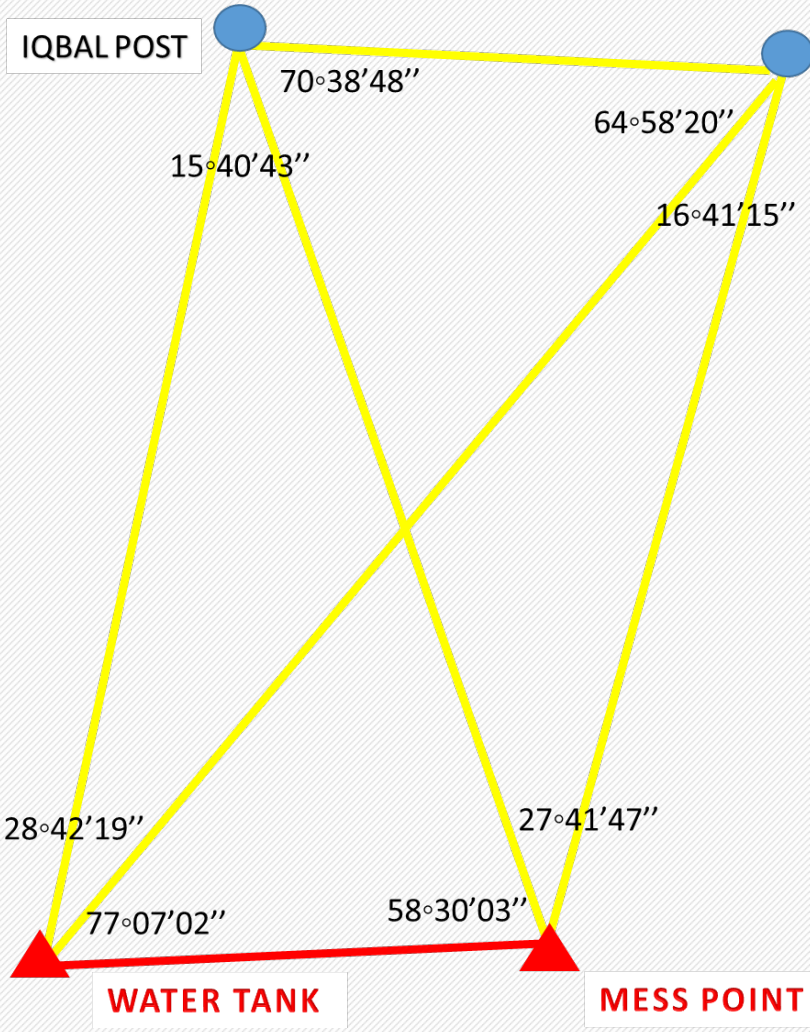
Step 1 : Angle Adjustment

Step 1 : Angle Adjustment

Iteration 1

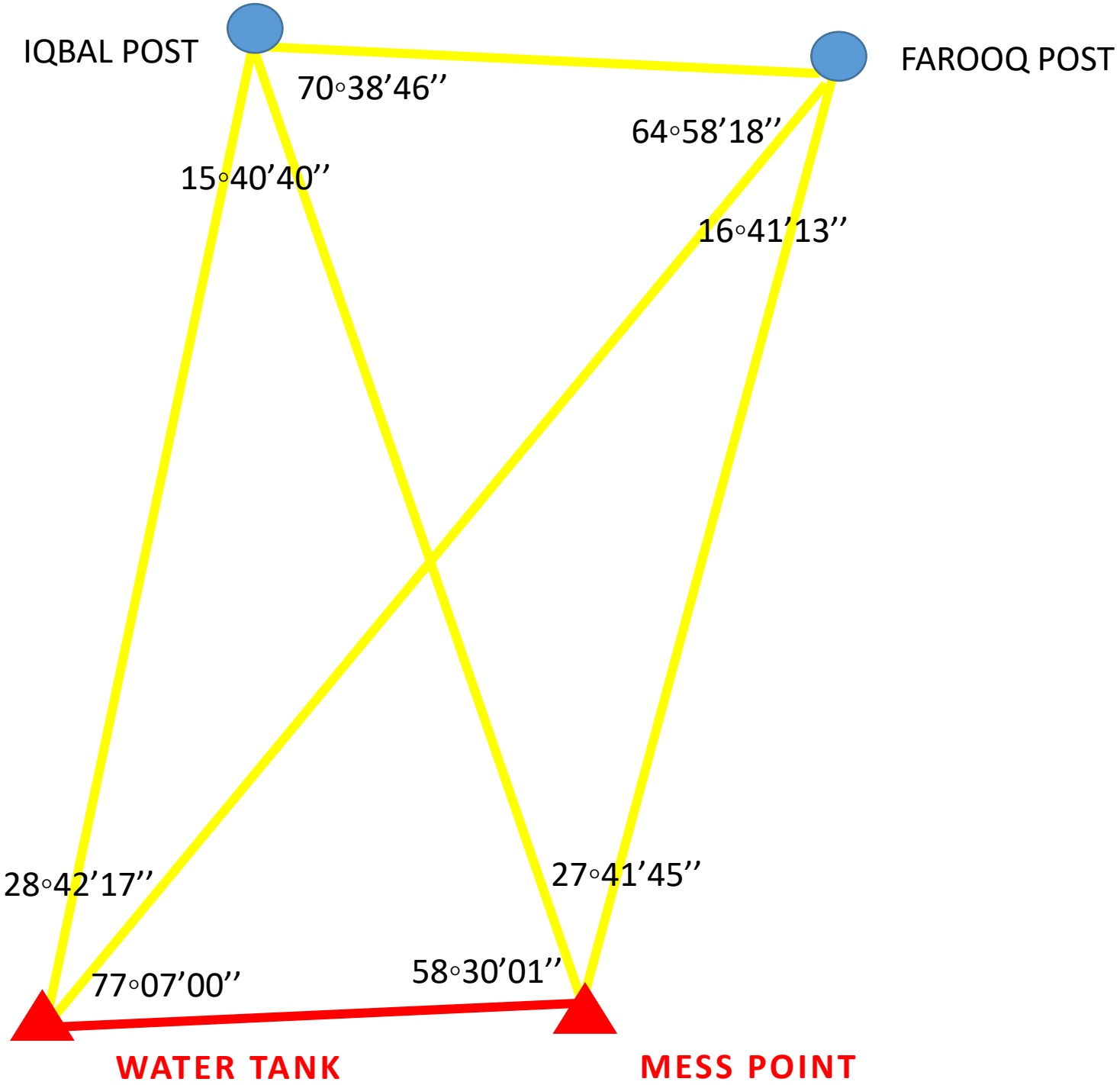
Check 1

S. No.	Observed Angles	Correction	Corrected Angles Check 1
1	70◦38’48”	-2”	70◦38’46”
2	64◦58’20”	-2”	64◦58’18”
3	16◦41’15”	-2”	16◦41’13”
4	27◦41’47”	-2”	27◦41’45”
5	58◦30’03”	-2”	58◦30’01”
6	77◦07’02”	-2”	77◦07’00”
7	28◦42’19”	-2”	28◦42’17”
8	15◦40’43”	-3”	15◦40’40”
Σ Observed Internal Angles =360◦00’17”			Σ Internal Angles = 360◦



Error=17”

Error per station = 17”/8


IQBAL POST
FAROOQ POST
70◦38’46’’
64◦58’18’’
15◦40’40’’
16◦41’13’’
28◦42’17’’
27◦41’45’’
77◦07’00’’
58◦30’01’’
WATER TANK
MESS POINT

Step 1 : Angle Adjustment

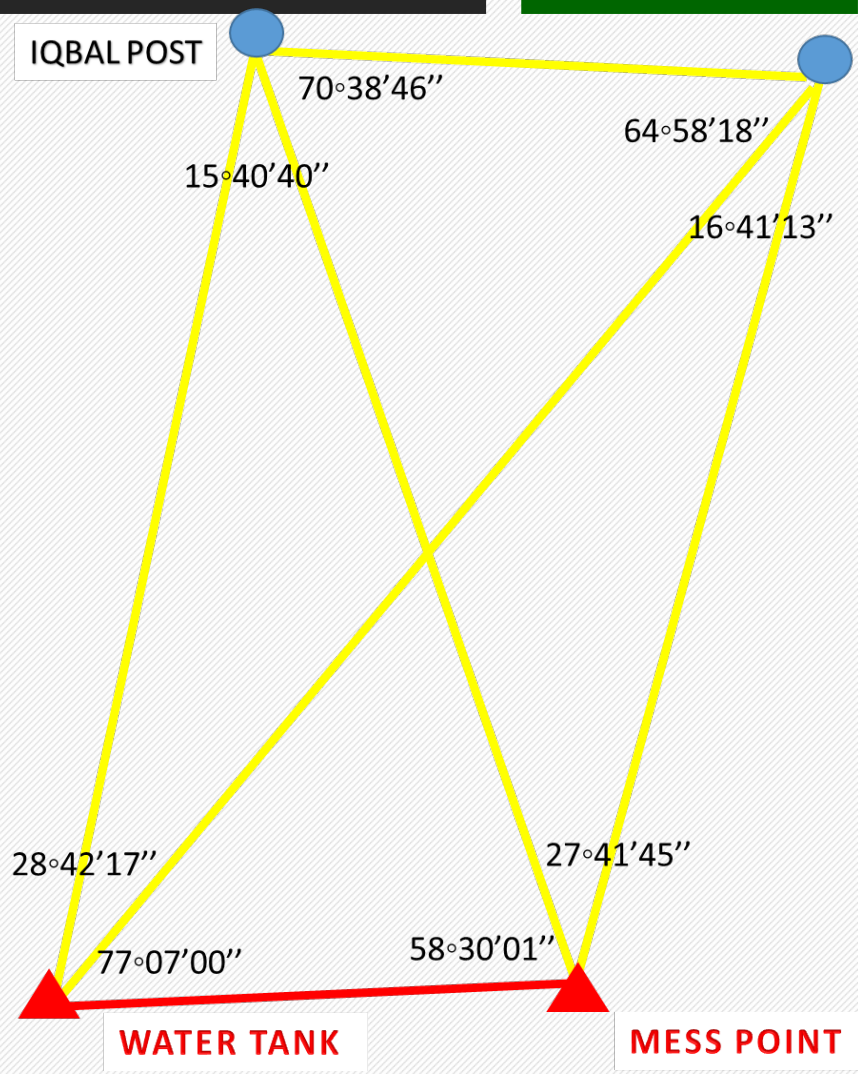
Step 1 : Angle Adjustment

Iteration 1

Check 2

Adjustment 1
Angle 1 + Angle 2 = Angle 6 + Angle 5
70◦38’46’’+ 64◦58’18’’= 77◦07’00’’+ 58◦30’01’’
135◦37’04’’ = 135◦37’01’’
Error = 03’’
Error per angle = 03’’/4

Since,
L.H.S > R.H.S
Subtract 01’’ from 2 angles in L.H.S
Add 01’’ in any 1 angle in L.H.S



Step 1 : Angle Adjustment

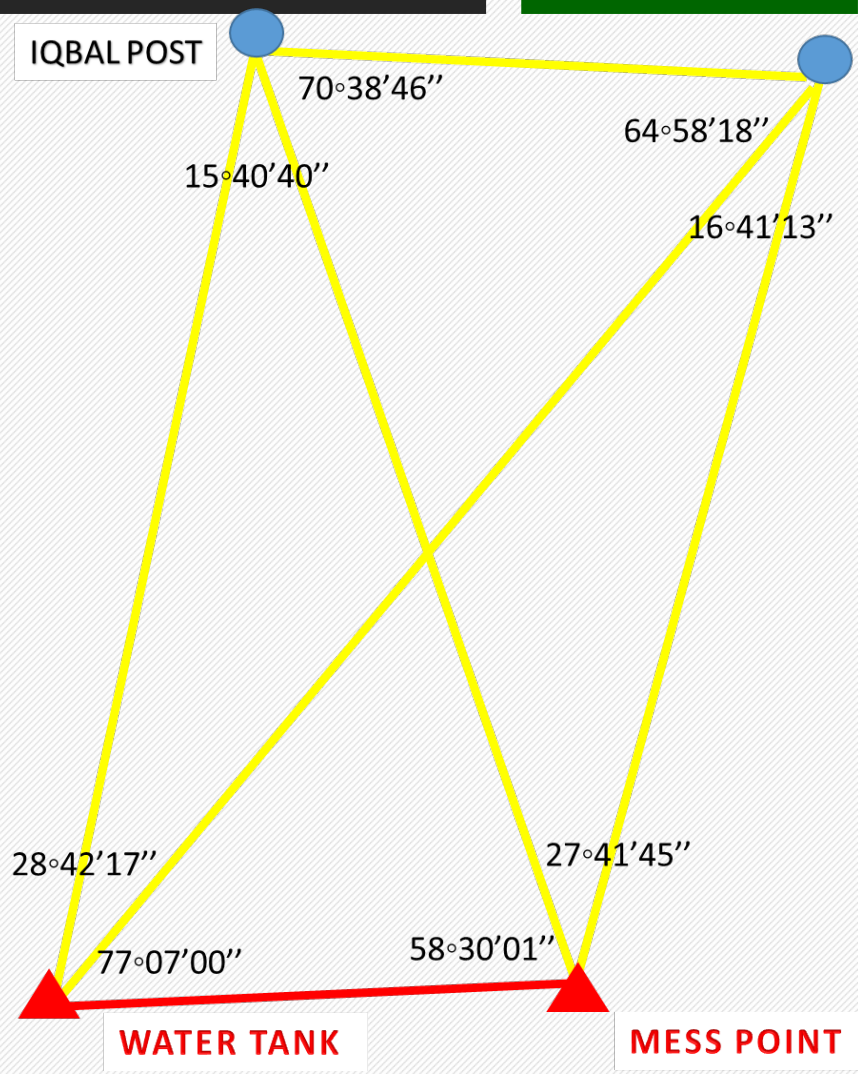
Step 1 : Angle Adjustment

Iteration 1

Check 2

Adjustment 2
Angle 8 + Angle 7 = Angle 3 + Angle 4
15◦40’40’’+ 28◦42’17’’= 16◦41’13’’+ 27◦41’45’’
44◦22’57’’ = 44◦22’58’’
Error = 01’’
Error per angle = 01’’/4

Since,
L.H.S < R.H.S
Subtract 01’’ from any 1 angle in R.H.S



Step 1 : Angle Adjustment

Step 1 : Angle Adjustment

Iteration 1

Check 2

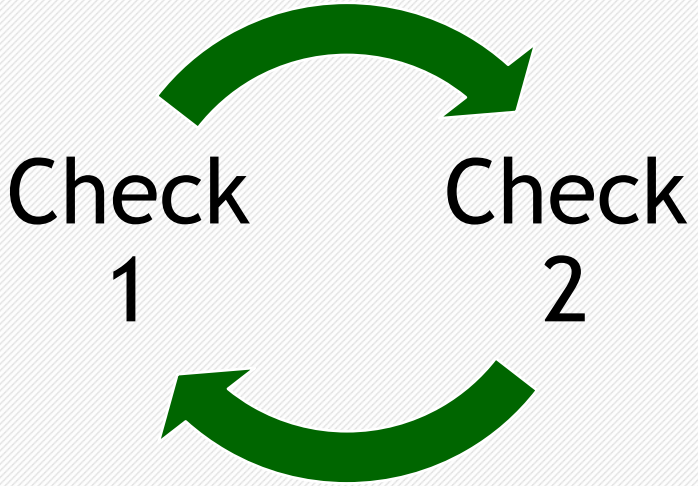
S. No.	Observed Angles	Correction	Corrected Angles Check 1
1	70◦38’46”	-1”	70◦38’45”
2	64◦58’18”	-1”	64◦58’17”
3	16◦41’13”	-1”	16◦41’12”
4	27◦41’45”		27◦41’45”
5	58◦30’01”	+1”	58◦30’02”
6	77◦07’00”		77◦07’00”
7	28◦42’17”		28◦42’17”
8	15◦40’40”		15◦40’40”
Σ Observed Internal Angles = 360◦00’00”		Apply Check 1 again Σ Internal Angles = 359◦59’58”	



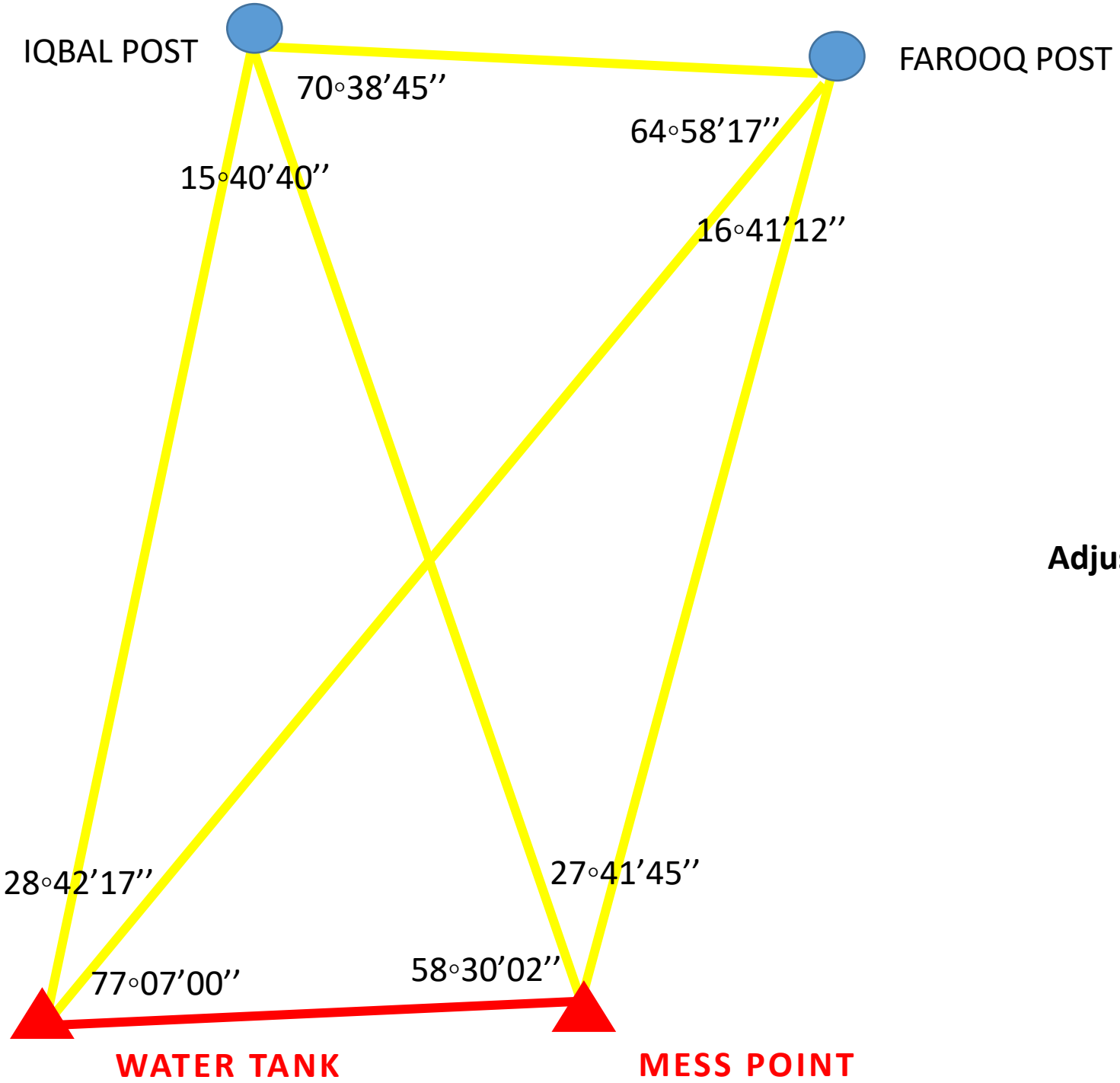


Step 1 : Angle Adjustment

ITERATIVE PROCESS







Adjustment from iteration 1

Step 1 : Angle Adjustment

Step 1 : Angle Adjustment

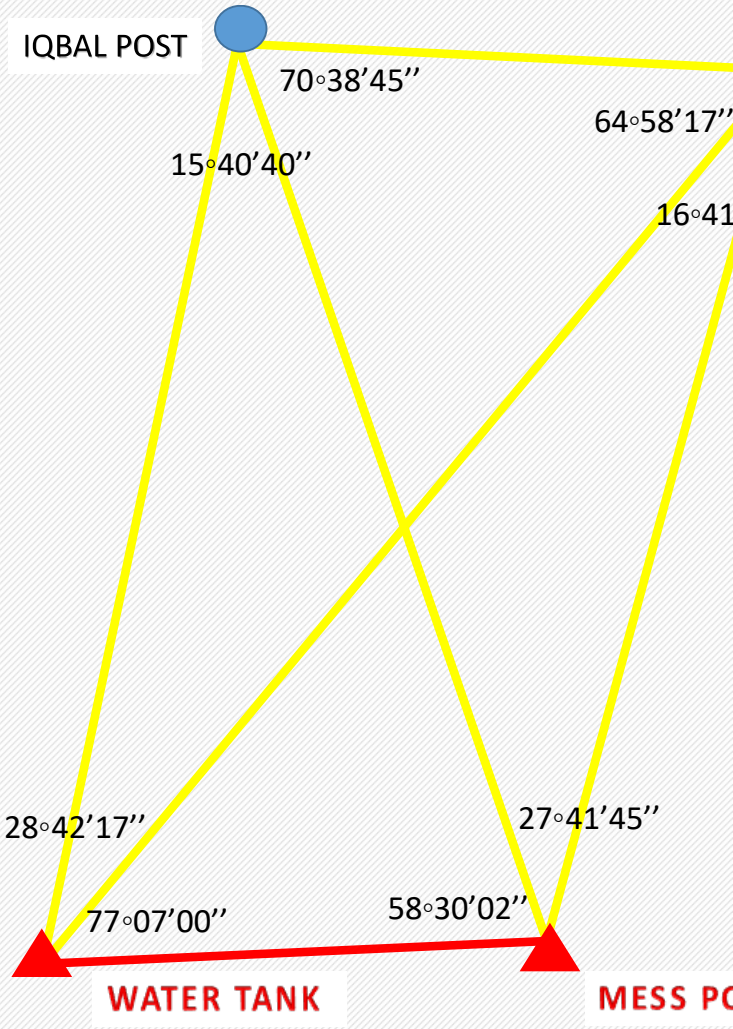
Iteration 2

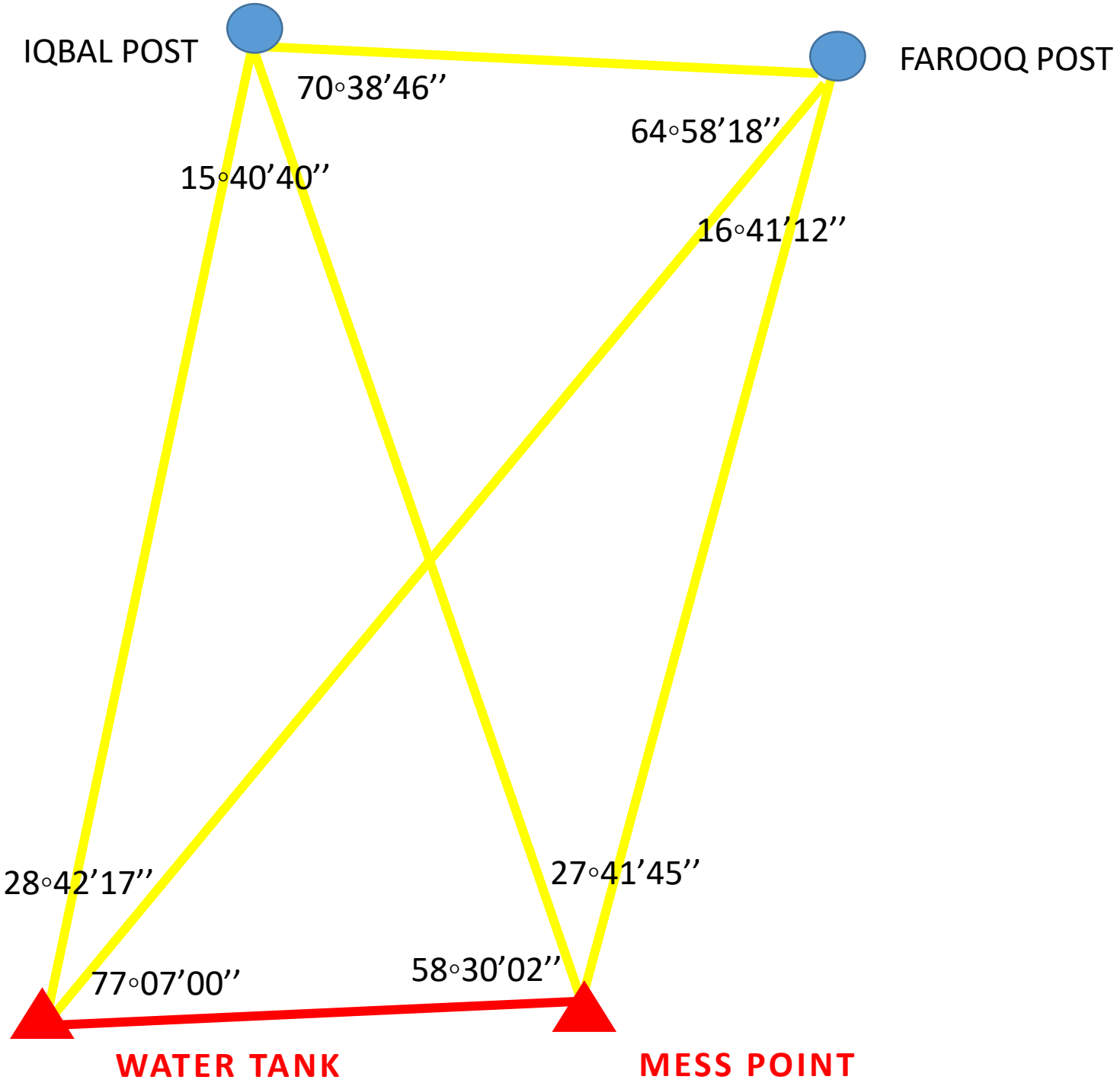
Check 1

S. No.	Observed Angles	Correction	Corrected Angles Check 1
1	70◦38’45”	+1”	70◦38’46”
2	64◦58’17”	+1”	64◦58’18”
3	16◦41’12”		16◦41’12”
4	27◦41’45”		27◦41’45”
5	58◦30’02”		58◦30’02”
6	77◦07’00”		77◦07’00”
7	28◦42’17”		28◦42’17”
8	15◦40’40”		15◦40’40”
Σ Observed Internal Angles = 359◦59’58”		Σ Internal Angles = 360◦	

Error=2”

Error per station = 2”/8




IQBAL POST
FAROOQ POST
70◦38’46’’
64◦58’18’’
15◦40’40’’
16◦41’12’’
28◦42’17’’
27◦41’45’’
77◦07’00’’
58◦30’02’’
WATER TANK
MESS POINT

Step 1 : Angle Adjustment

Step 1 : Angle Adjustment

Iteration 2

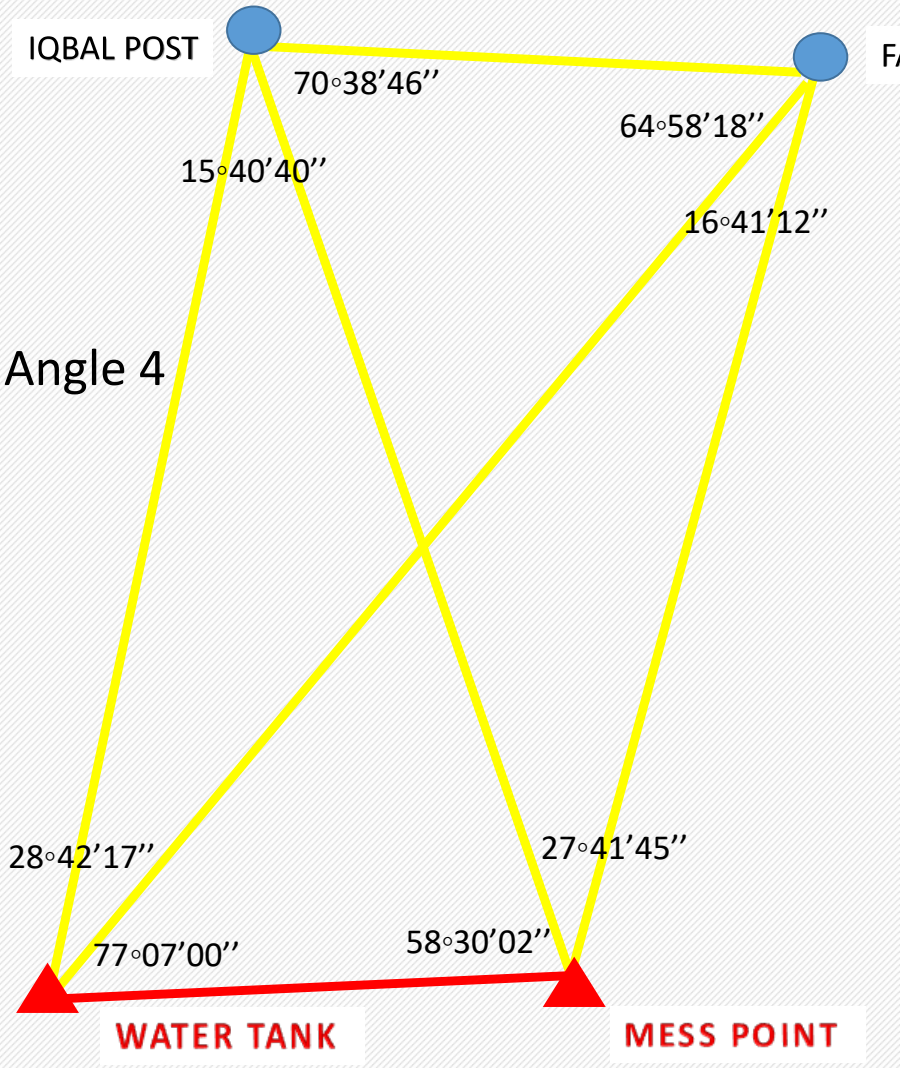
Check 2

Adjustment 1
Angle 1 + Angle 2 = Angle 6 + Angle 5
135◦37’04’’ = 135◦37’02’’
Error = 02’’
Error per angle = 02’’/4

L.H.S > R.H.S
Subtract 01’’ from angles in L.H.S
Add 01’’ in angles in L.H.S

Adjustment 2
Angle 8 + Angle 7 = Angle 3 + Angle 4
44◦22’57’’ = 44◦22’57’’

OK!



Step 1 : Angle Adjustment

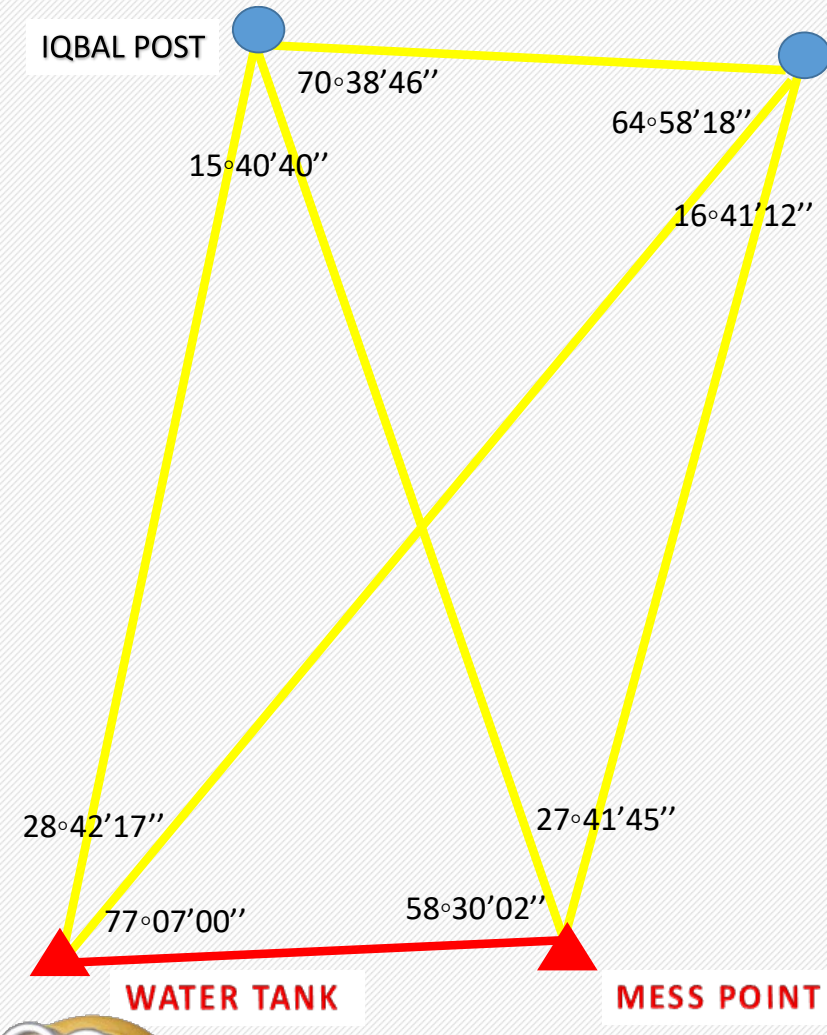
Step 1 : Angle Adjustment

Iteration 2

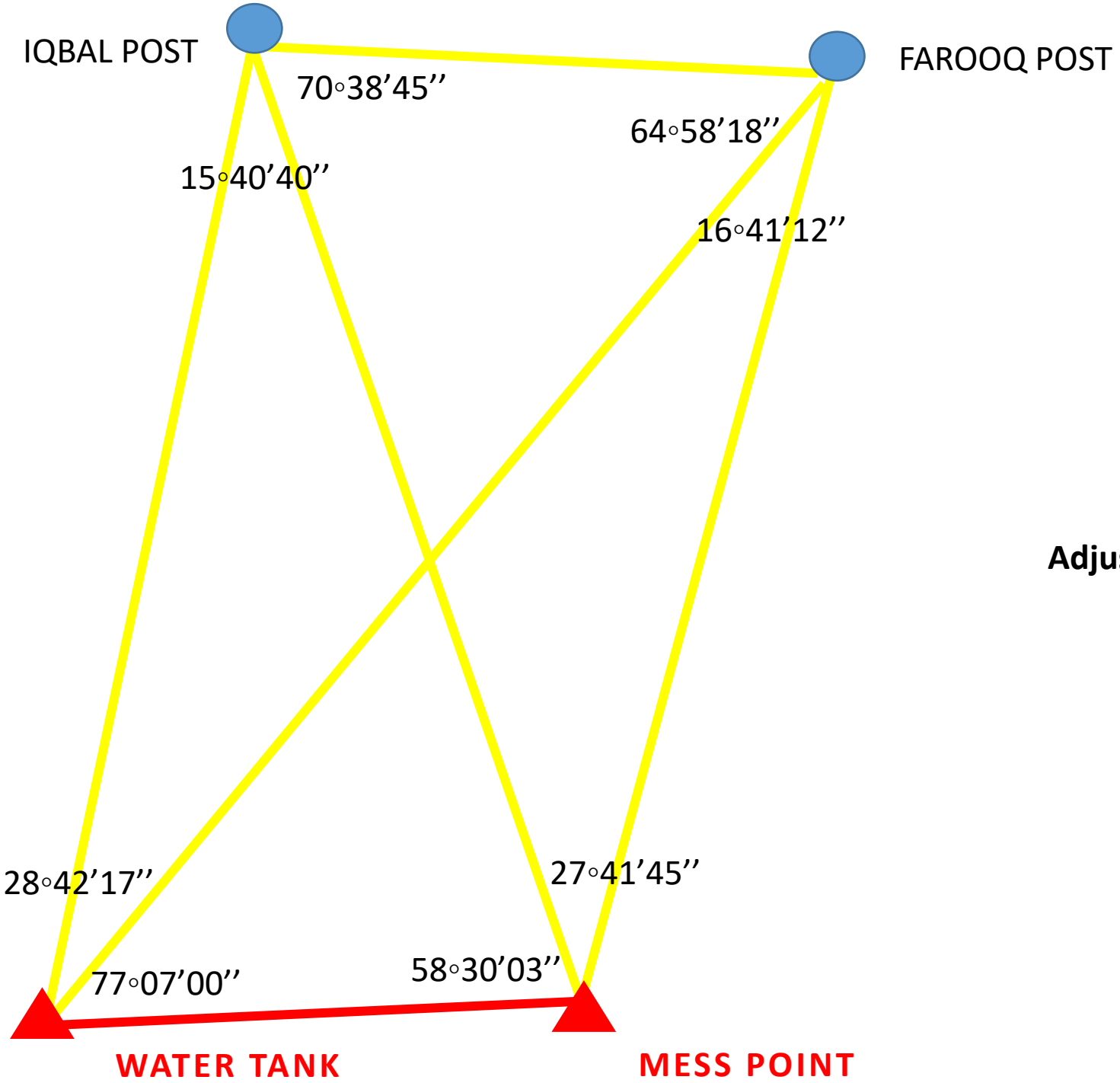
Check 2

S. No.	Observed Angles	Correction	Corrected Angles Check 1
1	70◦38’46”	-1”	70◦38’45”
2	64◦58’18”		64◦58’18”
3	16◦41’12”		16◦41’12”
4	27◦41’45”		27◦41’45”
5	58◦30’02”	+1”	58◦30’03”
6	77◦07’00”		77◦07’00”
7	28◦42’17”		28◦42’17”
8	15◦40’40”		15◦40’40”
Σ Observed Internal Angles = 359◦59’58”			

Apply Check 1 again
Σ Internal Angles = 360◦00’00”
OK!







Adjustment from iteration 2

Step 1 : Angle Adjustment

Step 1 : Angle Adjustment

Check 3

∑ Internal Angles of Triangles = 180◦

Consider triangles with base line

Triangle 1

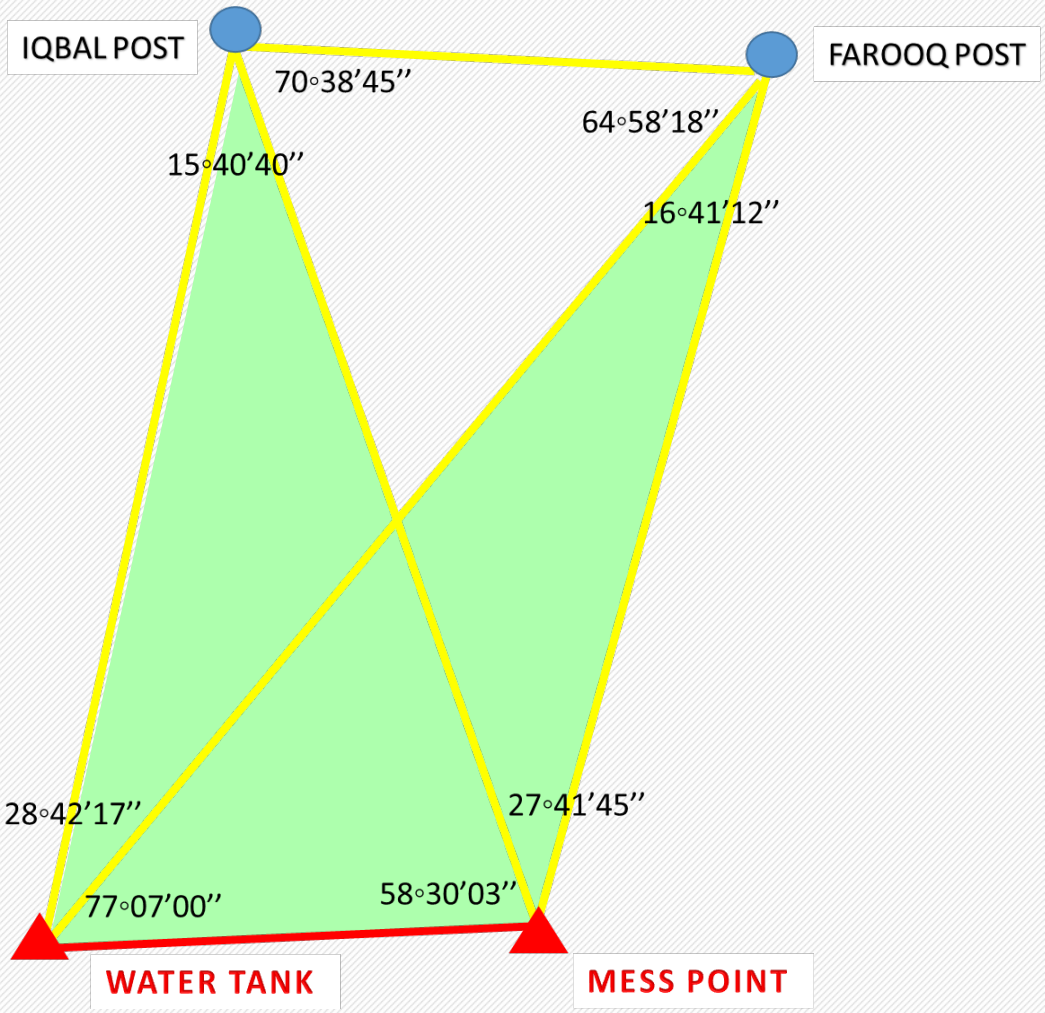
∑ Internal Angles of Triangles = 180◦

OK!

Triangle 2

∑ Internal Angles of Triangles = 180◦

OK!



Computations

Step 2 : Bearings

Consider 1 Triangle at a time

$$\text{Tan } \theta = \frac{\Delta E}{\Delta N}$$

$$\text{Tan } \theta = \frac{10517755.20 - 10516362.65}{3448679.37 - 3448359.91}$$

$\theta = 77^\circ 4' 48''$

Bearing WT to MP = 77◦04’48’’

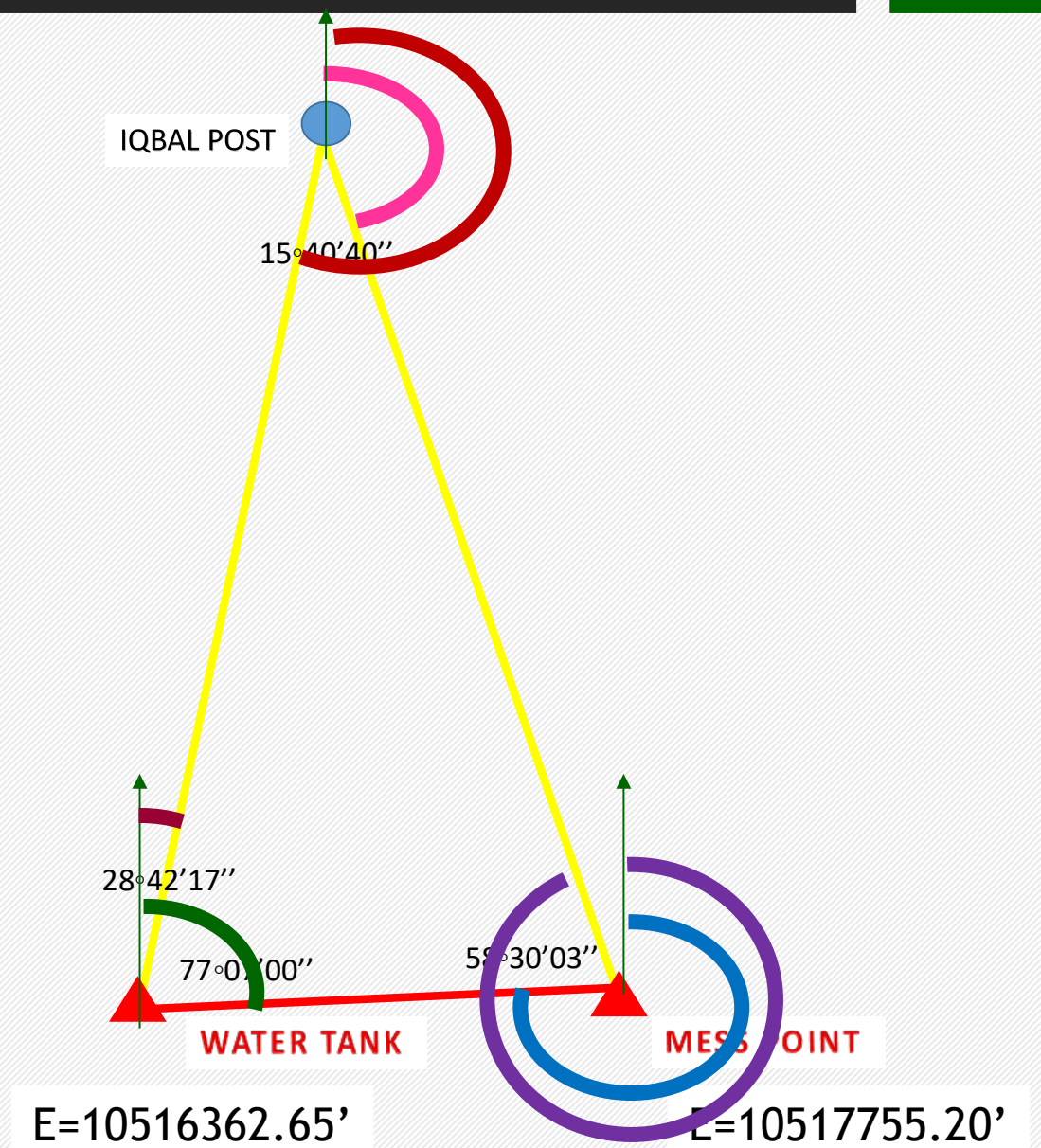
Back Bearing = 257◦04’48’’

Bearing MP to IP = 315◦34’51’’

Back Bearing = 135◦34’51’’

Bearing IP to WT = 151◦15’31’’

Back Bearing = 331◦15’31’’



Computations

Step 3 : Distances

Consider 1 Triangle at a time

Find ‘a’ using distance formula.

$a = \sqrt{(Ea - Eb)^2 + (Na - Nb)^2}$

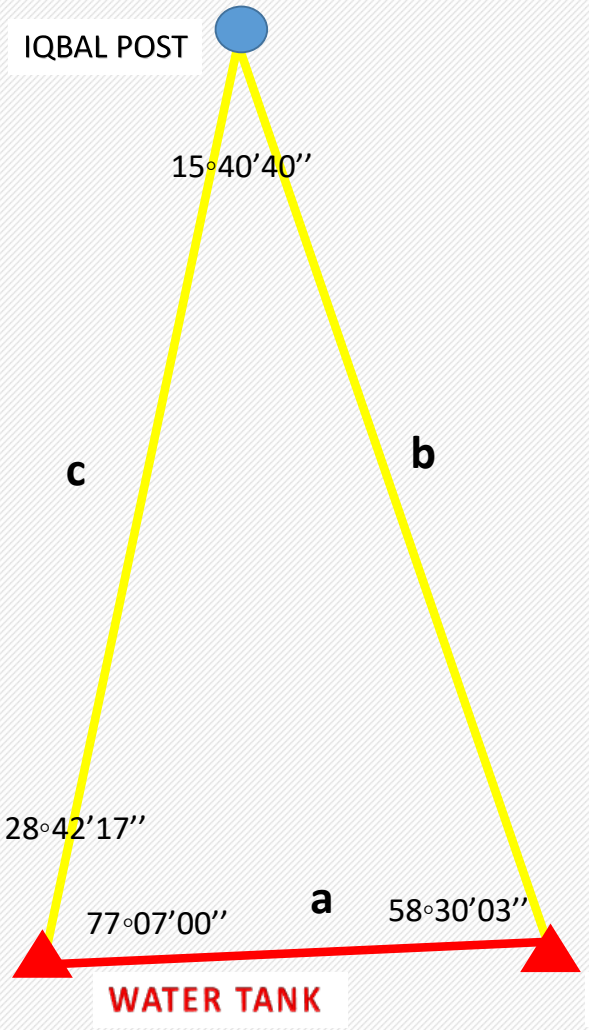
a = 1428.72’

Sine Law

$$\frac{a}{sin\alpha} = \frac{b}{sin\beta} = \frac{c}{sin\gamma}$$

b = 4507.94’

c = 5086.72’



E=10516362.65’
N=3448359.91’

E=10517755.20’
N=3448679.36’

Computations

Step 4 : Co-ordinates

Consider 1 Triangle at a time

Iqbal Post

From Water Tank

Easting

$E_{IP} = E_{WT} + D\sin\theta_{WT\ to\ IP}$

$E_{IP} = 10516362.65 + 4508.04\sin(331°15'31'')$

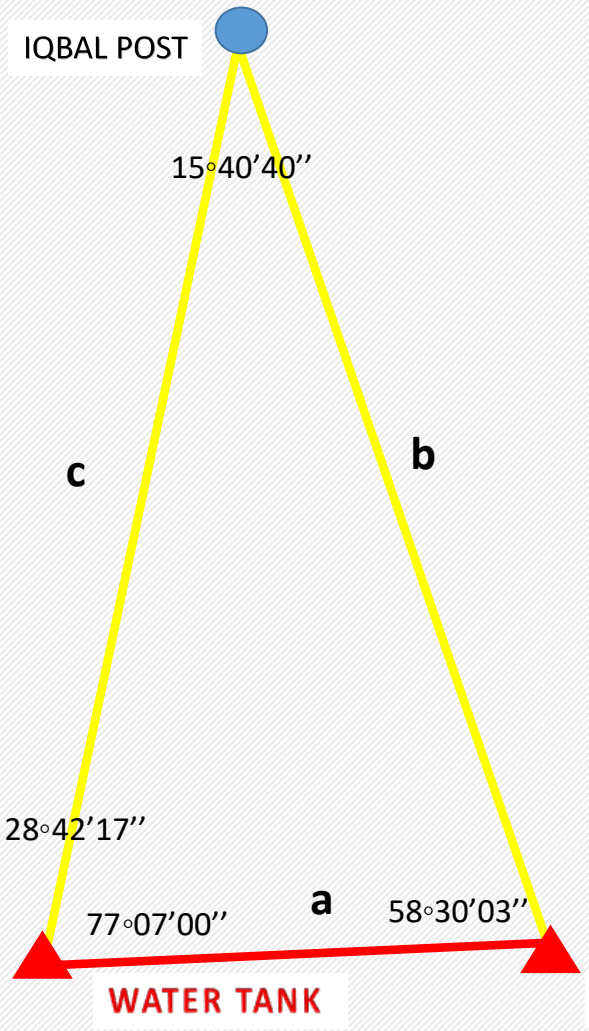
$E_{IP} = 10514194.93'$

Northing

$N_{IP} = N_{WT} + D\cos\theta_{WT\ to\ IP}$

$N_{IP} = 3448359.91 + 4508.04\cos(331°15'31'')$

$N_{IP} = 3452312.56'$



E=10516362.65'
N=3448359.91'

E=10517755.20'
N=3448679.36'

Computations

Step 4 : Co-ordinates

Consider 1 Triangle at a time

Iqbal Post

From Mess Point

Easting

$E_{IP} = E_{MP} + D\sin\theta_{MP\ to\ IP}$

$E_{IP} = 10517755.20 + 5086.81\sin(315°34'51'')$

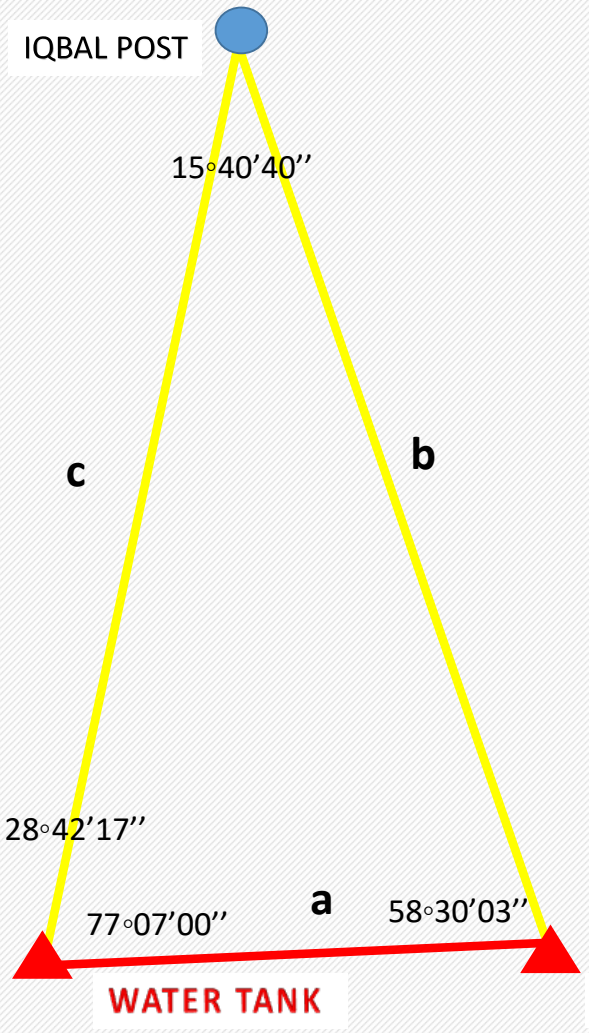
$E_{IP} = 10514194.93'$

Northing

$N_{IP} = N_{MP} + D\cos\theta_{MP\ to\ IP}$

$N_{IP} = 3448679.36 + 5086.81\cos(315°34'51'')$

$N_{IP} = 3452312.56'$



E=10516362.65'
N=3448359.91'

E=10517755.20'
N=3448679.36'

Computations

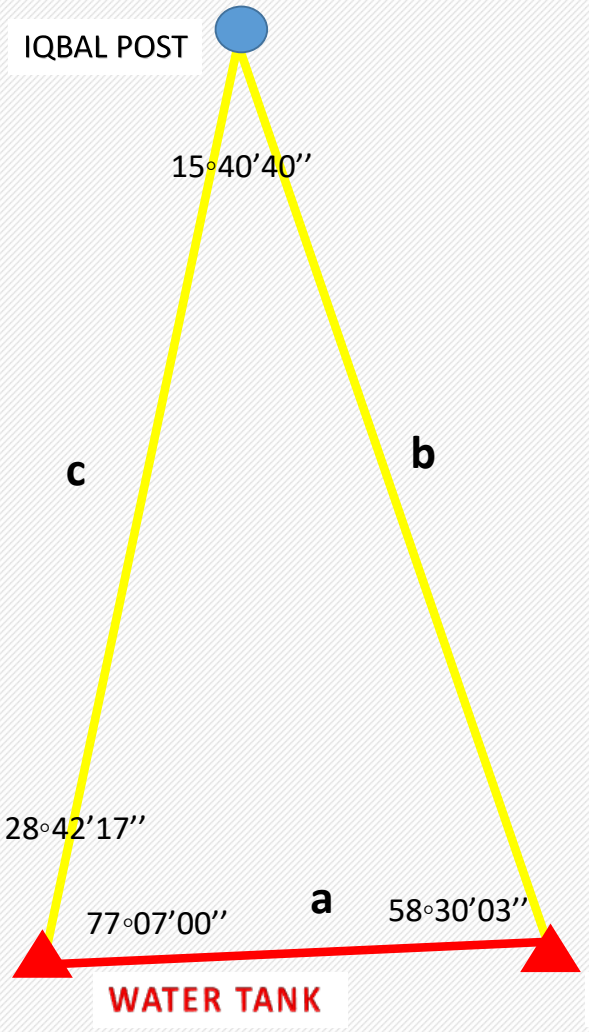
Step 4 : Co-ordinates

Consider 1 Triangle at a time

Iqbal Post

Mean Co-ordinates

Determine the mean of the pair of Eastings and Northings.

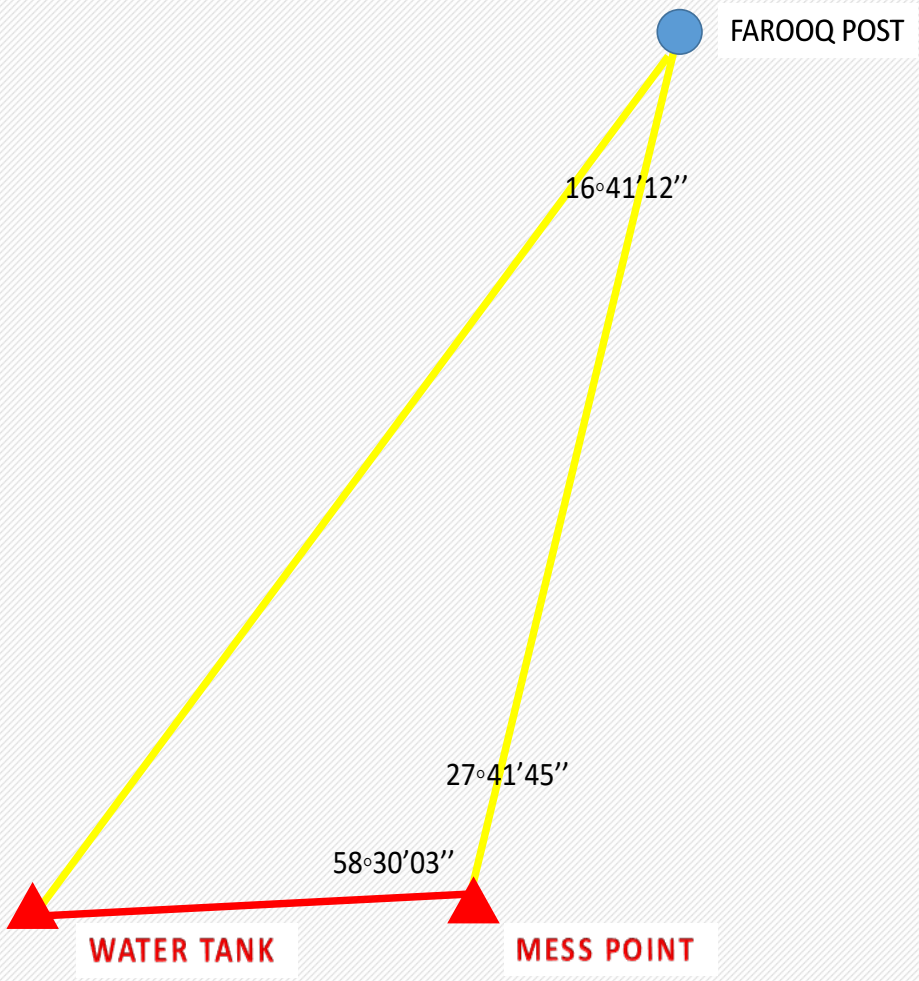


E=10516362.65’
N=3448359.91’

E=10517755.20’
N=3448679.36’

Computations

Repeat steps 2 to 4 for Triangle 2 to find the co-ordinates for Farooq Post.



E=10516362.65’
N=3448359.91’

E=10517755.20’
N=3448679.36’

Submissions

1. Report

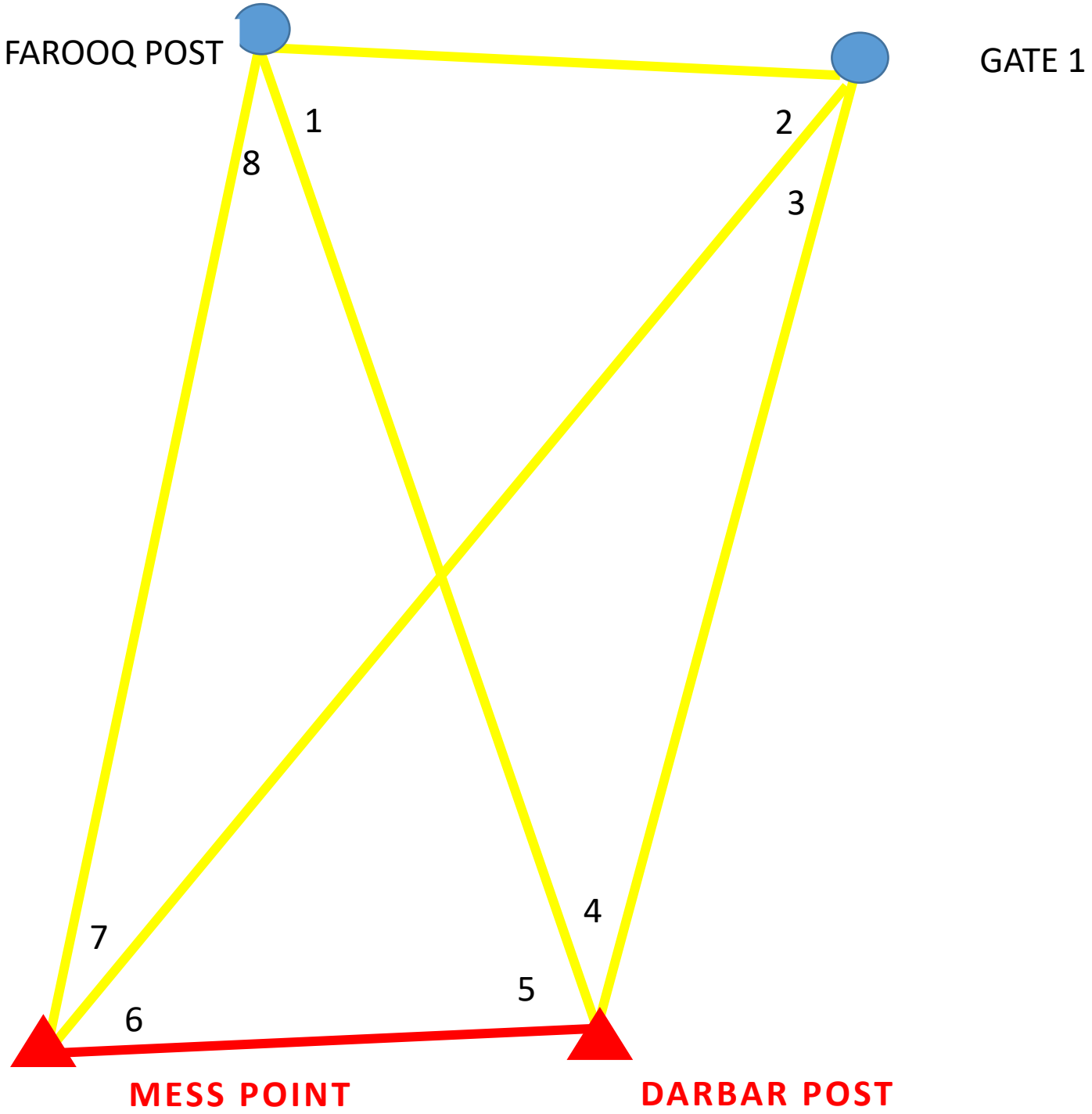
Summarize the following in a tabular form:

- Internal Angles
- Bearings
- Distances
- Co-ordinates

2. Calculations

Due Date:

Whenever you have the next FTF lab session.


FAROOQ POST
GATE 1
1
2
8
3
4
7
5
6
MESS POINT
DARBAR POST